$$\int_0^\infty x e^{-x}\,dx = \lim_{b\to\infty}\left[-e^{-x}(x+1)\right]_0^b \to -\frac{\overset{\uparrow\ \frac{\infty}{\infty} = -\frac{1}{e^x} = \frac{-1}{\infty} = \boxed{0}}{(x+1)}}{e^x} + \frac{\overset{\to 1}{(x+1)}}{e^x} = ①$$

$$\int_0^\infty 2^{-x}\,dx = \lim_{b\to\infty}\left[\frac{-2^{-x}}{\ln 2}\right]_0^b = \overset{\to 0}{\frac{-1}{\underbrace{2^b}_{\to+\infty}\ln 2}} + \frac{1}{\ln 2} = \boxed{\frac{1}{\ln 2}}$$

$$\hookrightarrow (e^{\ln 2})^{-x} = e^{-x\ln 2} \quad \hookrightarrow a = -\ln 2 \quad \to e^{ax} \to \frac{1}{a}e^{ax}$$

$$\int_0^\infty e^{-x}\sin x\,dx = -\sin x\, e^{-x} - \int(-e^{-x})\cdot\cos x\,dx$$

$$= -\sin x\, e^{-x} - \cos x\, e^{-x} - \underbrace{\int e^{-x}\cdot\sin x\,dx}_{\text{le si torna qui}}$$

$$\hookrightarrow I$$

$$I = -e^{-x}(\sin x + \cos x) - I$$

$$2I = -e^{-x}(\sin x + \cos x)$$

$$I = -\frac{e^{-x}}{2}(\sin + \cos x) \xrightarrow{\text{quindi}} \int_0^\infty e^{-x}\sin x\,dx = \lim_{b\to\infty}\left[-\frac{e^{-x}}{2}(\sin x + \cos x)\right]_0^b = 0 - \left(-\frac{1}{2}\right) = \boxed{\frac{1}{2}}$$

$$\to \lim_{b\to\infty}\underbrace{\left(-\frac{e^{-b}}{2}\right)}_{\to 0}(\sin b + \cos b) = 0$$

$$\to -\frac{e^{0}}{2}(\sin(0) + \cos(0)) = -\frac{1}{2}(0+1) = -\frac{1}{2}$$

$$\int_0^1 \frac{1}{1-x^2}\,dx = \int_0^1 \frac{1}{(1+x)(1-x)}\,dx = \frac{1}{2}\int_0^1\left(\frac{1}{1-x} + \frac{1}{1+x}\right)dx$$

$$\int_0^1 \frac{A}{(1-x)} + \frac{B}{(1+x)} \qquad = \frac{1}{2}\Big[\underbrace{-\ln|1-x|}_{\text{Converge}} + \ln|1+x|\Big]_0^1$$

$$1 = A(1+x) + B(1-x) \qquad = +\infty$$

Se $x=1$: $1 = 2A \to A = 1/2$

Se $x=-1$: $1 = 2B \to B = 1/2$

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} = 2\int_0^\infty \frac{1}{1+x^2} = 2\lim_{b\to\infty}\left[\arctan(x)\right]_0^b =$$

$$= 2\cdot\frac{\pi}{2} = \pi$$

$$\int_0^\infty \frac{x}{1+x^3}\,dx$$

$$\hookrightarrow \frac{A}{1+x} + \frac{Bx+C}{1-x+x^2} \cdot \qquad x = A(1-x+x^2) + (Bx+C)(x+1)$$

Se $x=-1$: $-1 = A(3) + (\cancel{-B+C})\cancel{(0)}$

$A = -1/3 \quad \to 0 = A+B = B = -A = \frac{1}{3}$

Se $x=0$: $0 = A + C$

$C = 1/3$

$$\frac{x}{1+x^3} = \frac{-1/3}{1+x} + \frac{1/3\,x + 1/3}{1-x+x^2}$$

$$\hookrightarrow \int\frac{x}{1+x^3} = -\frac{1}{3}\int\frac{1}{1+x}\,dx \quad \frac{1}{3}\int\frac{x+1}{1-x+x^2}\,dx$$

$\downarrow$

modifico il numeratore

$$\frac{1}{2}\int\frac{2x-1+3}{1-x+x^2}\,dx = \frac{1}{2}\ln|x^2-x+1| +$$

$$+ \frac{3}{2}\int\frac{1}{(x-1/2)^2 + 3/4}\,dx$$

$$= \frac{1}{2}\ln|x^2-x+1| + \sqrt{3}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)$$

Con $x\to\infty$

$$F(x) = \underbrace{\frac{1}{6}\ln\left(\frac{x^2-x+1}{(x+1)^2}\right)}_{\to 0} + \underbrace{\frac{1}{\sqrt{3}}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)}_{\frac{1}{\sqrt{3}}\cdot\frac{\pi}{2}}$$

Con $x\to 0$

$$\underbrace{\qquad\qquad}_{\to 0} + \frac{1}{3}\cdot\arctan\left(-\frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}}\cdot\left(-\frac{\pi}{6}\right)$$

$$\frac{\pi}{2\sqrt{3}} \to \frac{2\pi}{3\sqrt{3}}$$